

You are already logged in as pfox27@gmail.com

Lessons

- Table of Contents
- Confidence Levels
- Notes
- Bookmarks
- Highlights

Evaluating Expressions

Evaluating Expressions

Toolbox



## Evaluating Expressions

An expression is written with numbers, variables, and operation symbols. If you are given the values of the variables, you can evaluate the expression. Evaluating an expression means to find its value.

### Example 1

Find the value of  $2x - y$  when  $x = 4$  and  $y = -3$ .

#### Step 1

Replace the variables with the given values.

$$2x - y$$

$$= 2(4) - (-3)$$

#### Step 2

Use the order of operations.

Multiply.  $2(4) - (-3)$

$$= 8 - (-3)$$

Add the opposite.  $= 8 + 3$

$$= 11$$

The value is **11**.

If multiple sets of brackets, work from the inside out!

$$\begin{aligned} & \frac{[3 - (4 - 1)]}{[8 + 9(2 + 3)]} = \frac{[3 - 3]}{[8 + 9(5)]} \\ & = \frac{0}{8 + 45} \\ & = \frac{0}{53} \\ & = 0 \end{aligned}$$

Hide

#### Rate Your Confidence

High

Medium

Low

Continue >>

#### Category

Lesson 2: Expressions

#### Related Questions:

[Practice questions taken related to this topic](#)

When you write a negative number, you are using a **negation** symbol ( $-$ ). The symbol means “the opposite of.” You can already see that  $-6$  is the opposite of 6. In the same way  $-x$  is the opposite of  $x$ , and  $-(n - 3)$  is the opposite of  $(n - 3)$ . Study how negation symbols are used in the following example.

### Example 2

If  $x = 5$ , which expression has the greater value,  $-(x)^2$  or  $(-x)^2$ ?

This expression squares 5 and then negates the result.

$$-(x)^2 = -(5)^2$$

$$= -(25)$$

$$= -25$$

This expression negates 5 and then squares the result.

$$(-x)^2 = (-5)^2$$

$$= 25$$

If  $x = 5$ , then  $(-x)^2$  is greater than  $-(x)^2$ .

$$(-x)^2 > -(x)^2$$

Hide

The next two examples show how to evaluate expressions that contain grouping symbols. Remember, absolute value bars and fraction bars are two kinds of grouping symbols.

### Example 3

Find the value of  $2|a - b| - a$  when  $a = 5$  and  $b = 8$ .

#### Step 1

Replace the variables with the given values.

$$2|a - b| - a$$

$$2|5 - 8| - 5$$

#### Step 2

Find the absolute value part of the expression.

$$2|5 - 8| - 5$$

$$2|-3| - 5$$

$$2(3) - 5$$

#### Step 3

Multiply; then subtract.

$$2(3) - 5$$

$$= 6 - 5$$

$$= 1$$

The value of the expression is **1**.

Hide

Compare these two expressions:

#### Expression 1

$$4x + 9 + 3x - 1$$

When  $x = 2$ , the expression is  $4(2) + 9 + 3(2) - 1 = 8 + 9 + 6 - 1 = 22$

Simplify:  $7x + 8 = 22$

#### Expression 2

$$7x + 8$$

When  $x = 2$ , the expression is  $7(2) + 8 = 22$

When  $x = 2$ , these two expressions have the same value. In fact, the expressions are equal for any value of the variable. The two expressions are called **equivalent expressions**. Expression 2 is a **simplified** form of Expression 1.

### Example 4

Simplify  $2x + 2 + x - 4$ .

#### Step 1

Group **like terms**. These are terms that have the same variable.

$$2x + 2 + x - 4$$

**Note:**  $x$  means the same as  $1x$ .

#### Step 2

Group the numbers. Use the rules for adding signed numbers.

$$3x + 2 - 4$$

#### Step 3

Write the simplified expression.

$$3x - 2$$

Hide

### Example 5

Simplify  $-6k + 3 + 2k - 7$ .

#### Step 1

Group the variables. Use the rules for adding signed numbers with the **coefficients** (the numbers multiplying the variable).

$$-6k + 3 + 2k - 7$$

#### Step 2

Group the numbers.

$$-4k + 3 - 7$$

#### Step 3

Write the simplified expression.

$$-4k - 4$$

Hide

Report Content Errors